Section B — Practical Application

Task 1: Workstation Assembly

Objective

To assemble a workstation for the startup office by installing the main hardware components, connecting the required cables, and verifying that the system passes the Power-On Self-Test (POST).

Required Components and Tools

- Computer cabinet (Chassis)
- Motherboard
- CPU (Central Processing Unit)
- CPU cooler and thermal paste
- RAM modules
- SSD or HDD
- Power Supply Unit (PSU)
- Monitor, keyboard, and mouse
- Screwdriver and PC tool kit

Component Installation Sequence

1. Prepare the Work Area

Ensure that the work area is clean and dry. Disconnect the power supply and use anti-static precautions to prevent damage to sensitive components.

2. Install the CPU

Open the CPU socket on the motherboard and carefully place the processor according to the alignment markings. Lock the socket and ensure that the CPU is properly seated.

3. Install the CPU Cooler

Apply thermal paste if required and attach the CPU cooler securely. Connect its fan cable to the CPU_FAN header on the motherboard.

4. Install RAM Modules

Open the RAM slot clips and insert compatible RAM modules. Apply even pressure until the modules are properly locked into place.

5. Install Storage Device

Install the SSD or HDD in the appropriate mounting location. Connect the SATA data and power cables for SATA drives, or secure an M.2 SSD in its supported motherboard slot.

6. Install the Motherboard

Place the motherboard inside the chassis using the correct standoffs. Secure it with screws without applying excessive pressure.

7. Install the Power Supply

Mount the PSU in the designated chassis area and secure it with screws.

8. Connect Internal Cables

Connect the required cables:

- 24-pin ATX motherboard power cable.
- 4-pin or 8-pin CPU power cable.
- SATA power and data cables, if using SATA storage.
- Front-panel power switch, reset switch, and LED connectors.
- CPU and chassis fan connectors.

9. Connect External Devices

Connect the monitor, keyboard, mouse, and power cable to the workstation.

10. Perform POST Verification

Switch on the computer and check whether it completes the Power-On Self-Test (POST). Verify that the system detects the CPU, RAM, and storage device and displays the BIOS/UEFI screen without hardware errors.

Expected Result

The workstation should be assembled correctly, all required cables should be connected, and the system should successfully complete POST without hardware errors.

Task 2: RAM Mismatch Troubleshooting

Objective

To identify and resolve a RAM mismatch error through BIOS/UEFI settings and confirm that the workstation boots normally after troubleshooting.

Possible Causes of RAM Mismatch

A RAM mismatch error can occur due to:

- Different RAM speeds, such as 2400 MHz and 3200 MHz.
- Incompatible memory timings.
- Different RAM capacities.
- Unsupported RAM generation or motherboard compatibility.
- Improperly seated RAM modules.

Troubleshooting Procedure

1. Identify the RAM Error

Start the computer and observe the BIOS/UEFI or POST messages. Check whether the system reports a memory configuration error or fails to detect the installed RAM correctly.

2. Enter BIOS/UEFI

Restart the computer and press the appropriate BIOS/UEFI key, such as Delete, F2, or F10, depending on the motherboard manufacturer.

3. Check Installed RAM

Open the memory information section and verify the detected RAM capacity, speed, and configuration. Compare these details with the installed RAM modules.

4. Identify the Root Cause

Check whether the RAM modules have different speeds, timings, capacities, or unsupported specifications. Also verify that the modules are installed in the recommended motherboard slots.

5. Resolve the Mismatch

- Use compatible RAM modules supported by the motherboard.
- Set the memory speed to a common supported speed if required.

- Configure compatible memory timings.
- Reseat the RAM modules after safely powering off and disconnecting the workstation.
- Disable an unstable memory overclocking profile if it is causing the issue.

6. Save Settings and Restart

Save the corrected BIOS/UEFI configuration and restart the computer.

7. Verify the System

Check whether the system detects the correct RAM capacity and completes POST successfully. Confirm that the operating system starts normally.

Expected Result

The RAM mismatch error should be resolved by correcting the memory configuration or reseating compatible RAM modules. The workstation should detect the installed memory and boot normally.

Task 3: LAN Topology Build

Objective

To build a Local Area Network (LAN) topology in Cisco Packet Tracer for five systems and one network printer using a Class C subnet and verify connectivity through ping tests.

Required Devices

- 1 Cisco 2960 Switch
- 5 PCs
- 1 Network Printer
- Copper Straight-Through cables

IP Addressing Plan

The network will use the following Class C private IP address range:

- Network Address: 192.168.1.0
- Subnet Mask: 255.255.255.0
- Default Gateway: Not required for communication within the same subnet.

Device	IP Address	Subnet Mask
PC0	192.168.1.10	255.255.255.0
PC1	192.168.1.11	255.255.255.0
PC2	192.168.1.12	255.255.255.0
PC3	192.168.1.13	255.255.255.0
PC4	192.168.1.14	255.255.255.0
Printer	192.168.1.20	255.255.255.0

Topology Setup

1. Add Network Devices

Open Cisco Packet Tracer and add one 2960 switch, five PCs, and one network printer to the workspace.

2. Connect the Devices

Use Copper Straight-Through cables to connect all five PCs and the printer to separate switch ports.

3. Configure IP Addresses

Open each PC and printer configuration and assign the IP address and subnet mask according to the addressing table.

4. Verify the Network

Wait for the switch connections to become active. Open the Command Prompt on a PC and use the ping command to test communication with the other devices.

Testing Commands

ping 192.168.1.11

ping 192.168.1.12

ping 192.168.1.13

ping 192.168.1.14

ping 192.168.1.20

These commands test connectivity from PC0 to the other PCs and the network printer.

Expected Result

All five PCs and the network printer should be connected to the switch and configured with unique IP addresses from the same Class C subnet. All devices should be reachable through successful ping tests.

Task 4: Clean OS Installation

Objective

To perform a clean installation of Windows or Linux using an ISO file, manually partition the disk, and optimize startup programs and services for faster booting.

Required Tools

- VirtualBox
- Windows or Linux ISO file
- Sufficient disk space
- Minimum required RAM for the selected operating system

Installation Procedure

1. Prepare the Installation Environment

Open VirtualBox and create a new virtual machine. Select the operating system type and version according to the ISO file.

2. Allocate Hardware Resources

Assign suitable RAM, CPU cores, and virtual disk capacity according to the operating system requirements.

3. Attach the ISO File

Open the virtual machine settings and attach the Windows or Linux ISO file as the installation media.

4. Start the Installation

Start the virtual machine and boot from the attached ISO file. Follow the installation instructions displayed on the screen.

5. Perform Manual Disk Partitioning

Select the manual or custom partitioning option instead of automatic or guided partitioning.

For a typical Linux installation, partitions may include:

- Root (/) partition for the operating system and applications.
- Swap space for additional virtual memory, if required.

- Home (/home) partition for user files, if separate storage is desired.

For Windows, use the custom installation option and manually create or select the required disk partitions. Allow the installer to create the necessary system and recovery partitions.

6. Complete OS Installation

Select the appropriate installation partition and continue the installation process. Configure the user account, password, region, and other initial settings.

7. Install Updates and Drivers

After installation, check for operating system updates and install the required drivers or VirtualBox Guest Additions where applicable.

8. Optimize Startup Programs and Services

Review startup applications and disable unnecessary programs that automatically launch during boot. Check background services and disable only those that are confirmed to be unnecessary.

Do not disable essential security, networking, or operating system services.

9. Restart and Verify

Restart the virtual machine and confirm that the operating system boots without errors. Check that the partitions are available and startup performance is improved.

Expected Result

The operating system should be installed successfully using an ISO file and manual disk partitioning. Unnecessary startup programs and services should be optimized, and the system should boot without errors.